

Neil Ghugare

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### **Final Project Idea**

Since I work in computational astrophysics, I was thinking of doing a project related to computational fluid dynamics (I don't intend to do any project directly related to my research). I plan on creating a fluid dynamics solver using the Lattice Boltzmann Method (D2Q9 LBM). I also think it would be interesting and very useful for my long-term career to learn how to do this on C++/CUDA with a GPU instead of a CPU, if possible, which I have personal access to (so it wouldn't be an issue where I wouldn't have a GPU). I plan on having a simulation that treats a fluid as a collection of particle distributions undergoing local collisions and streaming steps. This would allow for a real-time simulation of complex wind flow and thermal diffusion of some geometries. Test cases/systems I could simulate would include a droplet relaxing to a sphere, the Poiseuille flow (fluid moving between two parallel plates), or a vortex like the Von Kármán vortex (turbulence of a wind).